

## MUMBAI SHIVAJI INTL, India

WMO: 430030

Lat: 19.089N

Lon: 72.868E

Elev: 37

StdP: 14.68

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | 62.9       | 65.9 | 42.4                          | 40.0 | 79.0 | 46.2 | 46.2 | 80.1 | 14.1                  | 80.4 | 13.1 | 81.5 | 1.4                      | 50   | 0.377 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 5             | 9.0                    | 96.8            | 72.9 | 95.0 | 73.4 | 93.4 | 74.1 | 82.4                | 88.2 | 81.7 | 87.5 | 81.0 | 86.8 | 8.3                  | 300  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 80.8                            | 160.3 | 86.0 | 80.3 | 157.7 | 85.6 | 79.2 | 151.6 | 84.8 | 46.5          | 88.8 | 45.7 | 87.9 | 45.0 | 87.4 | 87.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 16.4              | 14.6 | 13.6 | 59.2                       | 102.5 | 4.1                | 2.4 | 56.2                                               | 104.3 | 53.7       | 105.7 | 51.4       | 107.1 | 48.4       | 108.9 |
|                   |      |      | 51.7                       | 84.1  | 3.0                | 1.2 | 49.6                                               | 85.0  | 47.8       | 85.7  | 46.2       | 86.4  | 44.0       | 87.3  |

## Monthly Climatic Design Conditions

|                                                                               |                                                                               | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                               |                                                                               | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg                                                                         | 82.6   | 77.3  | 79.0  | 82.8  | 84.9  | 86.7  | 84.9  | 82.4  | 82.3  | 82.6  | 85.0  | 83.5  | 79.7 |      |
|                                                                               | DBStd                                                                         | 3.93   | 3.77  | 4.24  | 3.49  | 2.54  | 1.79  | 3.02  | 2.16  | 1.87  | 2.03  | 2.71  | 2.87  | 3.44 |      |
|                                                                               | HDD50                                                                         | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |      |
|                                                                               | HDD65                                                                         | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |      |
|                                                                               | CDD50                                                                         | 11903  | 847   | 813   | 1017  | 1046  | 1138  | 1047  | 1005  | 1001  | 977   | 1086  | 1005  | 921  |      |
|                                                                               | CDD65                                                                         | 6428   | 382   | 393   | 552   | 596   | 673   | 597   | 540   | 536   | 527   | 621   | 555   | 456  |      |
|                                                                               | CDH74                                                                         | 74766  | 3234  | 3782  | 6355  | 7559  | 9118  | 7663  | 6180  | 6017  | 5974  | 7807  | 6563  | 4514 |      |
|                                                                               | CDH80                                                                         | 31383  | 1132  | 1510  | 2856  | 3490  | 4680  | 3498  | 1975  | 1792  | 1961  | 3673  | 3040  | 1776 |      |
| Wind                                                                          | WSAvg                                                                         | 6.1    | 4.4   | 4.8   | 5.5   | 5.9   | 7.2   | 8.4   | 9.3   | 8.7   | 5.9   | 4.7   | 4.5   | 4.4  |      |
| Precipitation                                                                 | PrecAvg                                                                       | 94.5   | 0.0   | 0.0   | 0.0   | 0.0   | 1.1   | 19.7  | 34.4  | 21.0  | 14.5  | 3.1   | 0.4   | 0.1  |      |
|                                                                               | PrecMax                                                                       | 136.3  | 0.6   | 0.3   | 0.4   | 0.2   | 11.9  | 40.2  | 54.9  | 39.6  | 35.3  | 14.6  | 3.3   | 0.9  |      |
|                                                                               | PrecMin                                                                       | 58.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 2.9   | 8.6   | 6.8   | 3.3   | 0.0   | 0.0   | 0.0  |      |
|                                                                               | PrecStd                                                                       | 17.3   | 0.1   | 0.1   | 0.1   | 0.0   | 2.8   | 10.6  | 11.5  | 9.2   | 7.5   | 3.1   | 0.7   | 0.2  |      |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%                                                                          | DB     | 93.3  | 97.2  | 100.3 | 98.5  | 95.1  | 94.6  | 89.8  | 89.8  | 93.3  | 98.2  | 96.9  | 94.6 |      |
|                                                                               |                                                                               | MCWB   | 69.8  | 70.5  | 70.9  | 74.3  | 78.3  | 80.7  | 79.5  | 79.2  | 79.4  | 74.0  | 72.9  | 70.5 |      |
|                                                                               | 2%                                                                            | DB     | 90.1  | 93.5  | 96.5  | 95.0  | 93.0  | 92.1  | 88.1  | 88.0  | 89.8  | 95.3  | 94.9  | 91.7 |      |
|                                                                               |                                                                               | MCWB   | 68.7  | 70.0  | 70.9  | 75.1  | 79.5  | 79.8  | 79.6  | 78.9  | 78.6  | 74.0  | 72.1  | 70.1 |      |
|                                                                               | 5%                                                                            | DB     | 88.1  | 91.1  | 93.3  | 92.9  | 91.6  | 91.3  | 87.4  | 86.4  | 88.0  | 93.6  | 93.1  | 89.9 |      |
|                                                                               |                                                                               | MCWB   | 68.4  | 69.8  | 71.2  | 75.6  | 79.3  | 79.7  | 79.5  | 78.5  | 78.6  | 74.7  | 71.8  | 69.8 |      |
|                                                                               | 10%                                                                           | DB     | 86.1  | 88.0  | 91.1  | 91.1  | 91.0  | 89.7  | 86.1  | 85.8  | 86.4  | 91.7  | 91.3  | 88.1 |      |
|                                                                               |                                                                               | MCWB   | 68.1  | 69.1  | 71.8  | 76.1  | 79.0  | 79.6  | 79.2  | 78.4  | 78.2  | 75.7  | 72.2  | 69.3 |      |
|                                                                               | Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%   | WB    | 75.8  | 76.0  | 80.7  | 82.2  | 83.2  | 83.3  | 82.0  | 81.4  | 82.3  | 82.8  | 80.1 | 77.6 |
|                                                                               |                                                                               |        | MCDB  | 84.4  | 86.4  | 87.8  | 90.4  | 90.7  | 88.9  | 85.4  | 85.1  | 87.1  | 88.8  | 88.1 | 85.8 |
| 2%                                                                            |                                                                               | WB     | 72.8  | 74.2  | 78.9  | 80.5  | 81.9  | 82.3  | 81.2  | 80.4  | 80.9  | 81.5  | 78.8  | 75.5 |      |
|                                                                               |                                                                               | MCDB   | 83.6  | 84.4  | 85.9  | 88.1  | 89.4  | 88.1  | 84.9  | 84.3  | 85.5  | 87.8  | 86.7  | 85.1 |      |
| 5%                                                                            |                                                                               | WB     | 71.4  | 72.8  | 77.4  | 79.6  | 81.1  | 81.6  | 80.6  | 79.9  | 80.0  | 80.4  | 77.6  | 74.0 |      |
|                                                                               |                                                                               | MCDB   | 82.7  | 83.7  | 85.4  | 87.2  | 88.7  | 87.6  | 84.6  | 83.9  | 84.6  | 87.1  | 86.0  | 83.9 |      |
| 10%                                                                           |                                                                               | WB     | 70.0  | 71.7  | 76.0  | 78.9  | 80.6  | 80.9  | 80.0  | 79.3  | 79.4  | 79.5  | 76.5  | 72.5 |      |
|                                                                               |                                                                               | MCDB   | 81.4  | 82.8  | 84.5  | 86.5  | 88.2  | 87.0  | 83.9  | 83.4  | 83.9  | 86.3  | 85.5  | 83.3 |      |
| Mean Daily<br>Temperature<br>Range                                            |                                                                               | 5% DB  | MDBR  | 19.2  | 18.5  | 16.4  | 12.5  | 9.0   | 8.1   | 6.8   | 6.7   | 8.7   | 13.8  | 17.7 | 19.0 |
|                                                                               |                                                                               |        | MCDBR | 20.7  | 22.2  | 20.5  | 15.6  | 9.9   | 9.3   | 8.1   | 7.8   | 10.7  | 16.9  | 19.1 | 20.2 |
|                                                                               | MCWBR                                                                         |        | 9.1   | 9.7   | 9.6   | 7.6   | 4.7   | 4.6   | 4.5   | 4.0   | 4.8   | 7.7   | 8.4   | 8.9  |      |
|                                                                               | 5% WB                                                                         | MCDBR  | 19.1  | 18.0  | 14.8  | 11.6  | 8.6   | 8.5   | 6.9   | 6.7   | 8.8   | 12.2  | 14.7  | 17.2 |      |
|                                                                               |                                                                               | MCWBR  | 9.1   | 9.2   | 8.5   | 6.8   | 4.7   | 5.0   | 4.5   | 3.9   | 4.7   | 6.5   | 7.0   | 9.1  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub                                                                          | 0.453  | 0.452 | 0.507 | 0.548 | 0.557 | 0.664 | 0.689 | 0.629 | 0.570 | 0.588 | 0.532 | 0.492 |      |      |
|                                                                               | taud                                                                          | 2.143  | 2.156 | 2.022 | 1.957 | 1.975 | 1.765 | 1.717 | 1.845 | 1.981 | 1.890 | 1.995 | 2.050 |      |      |
|                                                                               | Ebn at Noon                                                                   | 260    | 268   | 258   | 248   | 243   | 216   | 211   | 225   | 238   | 227   | 234   | 242   |      |      |
|                                                                               | Edh at Noon                                                                   | 46     | 48    | 56    | 61    | 59    | 72    | 75    | 67    | 58    | 61    | 53    | 49    |      |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg                                                                        | 1577   | 1849  | 2122  | 2279  | 2244  | 1504  | 1117  | 1197  | 1479  | 1663  | 1564  | 1493  |      |      |
|                                                                               | RadStd                                                                        | 50     | 48    | 61    | 42    | 99    | 164   | 165   | 100   | 131   | 90    | 77    | 52    |      |      |

## Historical Trends

|                        |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| Station Only           |  | +0.72 | +3.02   | -1.58  | N/A     | N/A   | +0.59 | N/A         | N/A   | +288  | +290  |
| Regional (2 neighbors) |  | +0.61 | +1.67   | N/A    | N/A     | N/A   | +0.40 | N/A         | N/A   | +288  | +290  |

Nomenclature: See separate page